

Simulation report - Experiment 2

Streaming CA-CFAR verification evidence

DUTs: cfar_core and radar_demo. Original SystemVerilog testbenches and exact source bytes match the successful Vivado build. Supplied local simulation: Icarus Verilog 12.0, 8 October 2026. Independent laptop confirmation: Vivado XSim 2025.1.1, same day.

Test suite	Observed pass output
tb_cfar_core	CORE SIMULATION PASS: checked=10721 accepted=12963
tb_radar_demo	DEMO SIMULATION PASS: 33 frame checks

Reference model: independent chronological sample array and expected-result queue. It checks full-width arithmetic, valid timing, index alignment and detection against each emitted DUT result. Coverage includes equality, training/guard exclusion, integer truncation, unsigned high values, display saturation, continuous/gapped traffic, random values/scales, short frames and reset.

Controller: four scenarios, four factors and two gap modes produce 32 combinations; one repeated standard frame gives 33 checks. The testbench additionally checks held-start behavior and reset during processing.

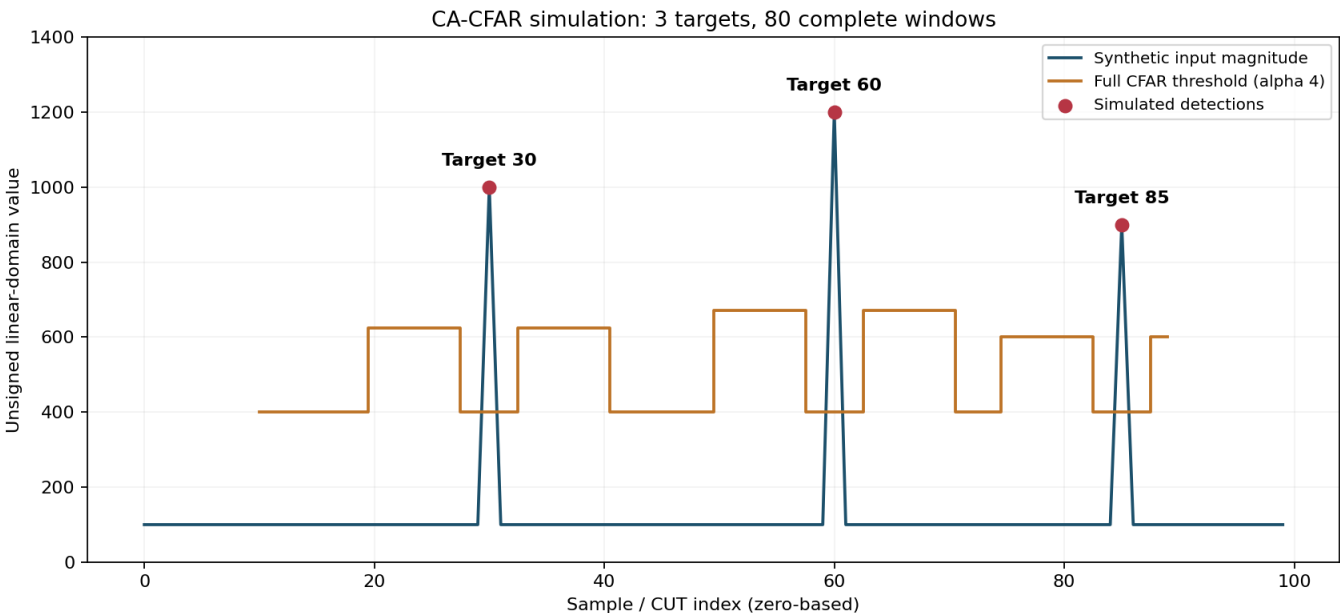


Figure S1. Plot from the original standard-frame result CSV. Exactly 80 full windows and three detections at 30, 60, 85; alpha four. Magnitudes are controlled synthetic unsigned linear-domain values, not RF measurements.

Waveform, timing and expected results

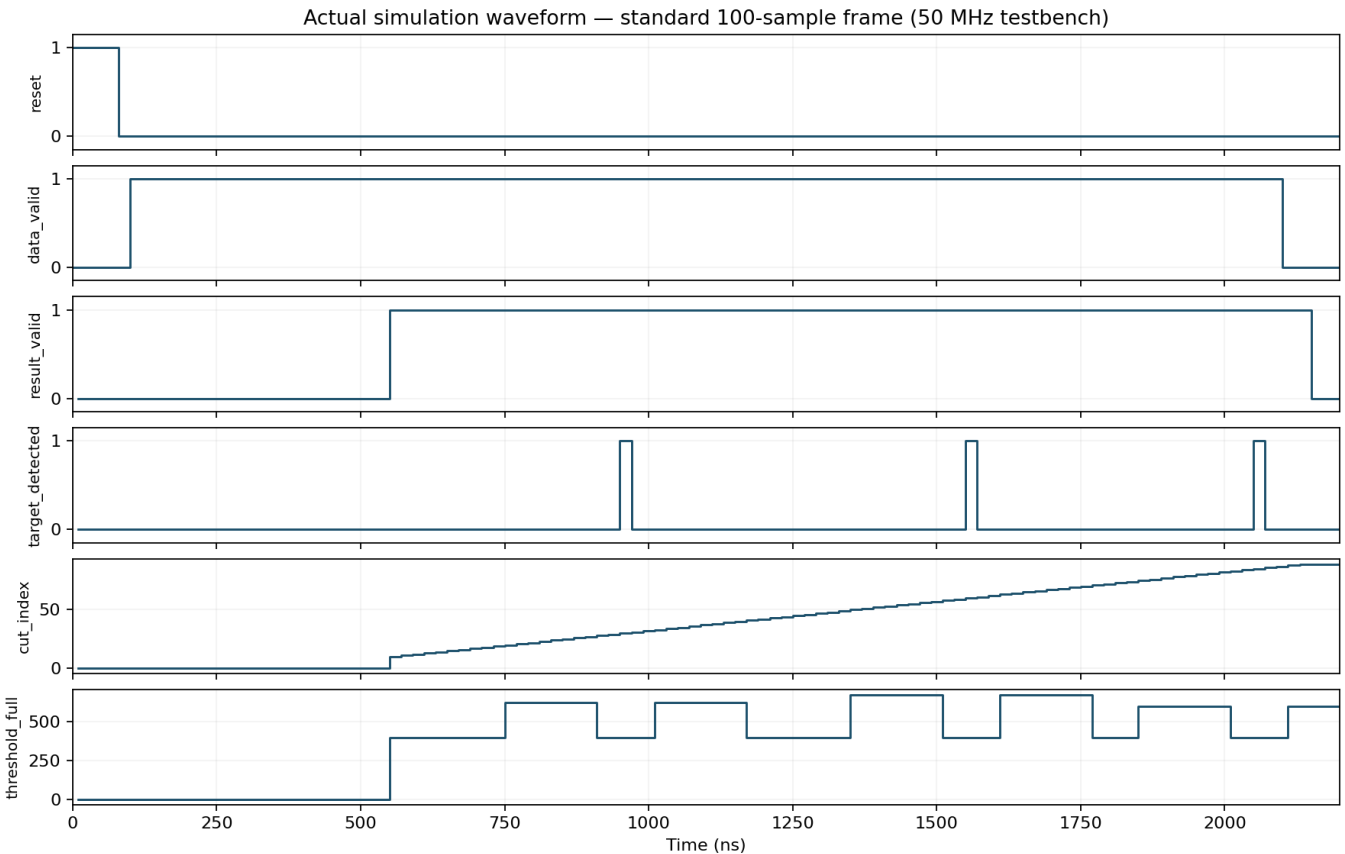


Figure S2. Real supplied cfar_core.vcd rendered as traces. Reset clears pending results; result_valid starts only after a full window and pipeline latency. Threshold_full is the comparison value, distinct from a saturating display. Full original traces are retained in Simulation/.

Case at alpha four	Detection result	Frame results
Constant 100	0 targets	80
1000/1200/900 at 30/60/85	3 targets: 30, 60, 85	80
Weak 300 at 30	0; factor two produces 1 at 30	80
20000 background / 60000 target	0: compare against full 80000 threshold	80

A result for CUT n-10 appears two clocks after the accepted newest sample n (n at least 20). Stalls change wall-clock spacing but never duplicate a sample window. The controller captures/drains outputs and finishes in 104 continuous cycles or 302 gapped cycles. At the confirmed 50 MHz design clock these equal 2.08 and 6.04 microseconds; host/JTAG time is excluded.

Transcripts and reproduction

The complete exact local output is Simulation/transcript.txt. The full successful XSim/synthesis/routing run is Evidence/vivado_build.log. Representative unmodified local transcript lines follow:

```
CORE SIMULATION PASS: checked=10721 accepted=12963
DEMO PASS sc=1 alpha=4 gap=0 detections=3 results=80 cycles=104
DEMO PASS sc=1 alpha=4 gap=1 detections=3 results=80 cycles=302
DEMO PASS sc=2 alpha=4 gap=0 detections=0 results=80 cycles=104
DEMO PASS sc=2 alpha=2 gap=0 detections=1 results=80 cycles=104
DEMO SIMULATION PASS: 33 frame checks
```

Reproduce locally with Icarus

Run from the experiment root with iverilog/vvp installed. The testbenches save VCD/CSV/PASS outputs. Use a separate copy to preserve this original archive.

```
mkdir -p Results Simulation
iverilog -g2012 -s tb_cfar_core -o core.vvp \
    RTL/cfar_core.v Testbench/tb_cfar_core.sv
vvp core.vvp +OUTDIR=.
iverilog -g2012 -s tb_radar_demo -o demo.vvp \
    RTL/cfar_core.v RTL/radar_demo.v Testbench/tb_radar_demo.sv
vvp demo.vvp +OUTDIR=.
```

For the complete vendor flow, RUN_PROJECT2.cmd invokes Vivado/build_project.tcl, which runs both XSim suites and requires their PASS markers before synthesis/routing. This flow recreates VIO and the native project. A new build clears old markers and hardware outputs; preserve the original archive first.

Schematic and evidence audit

The Yosys structural check generates the actual top schematic. Simulation/vio_elaboration_stub.v and vendor clock-library declarations serve as black boxes; this check does not verify their implementation, lock, routing or timing. Images contains the DOT/PNG/SVG; Simulation contains the used-module JSON and complete tool transcript. The generation recipe is Tools/SCHEMATIC_REPRODUCTION.md.

Run python Tools/validate_evidence.py for source/bitstream identity and saved CSV consistency. No simulation or new board run is claimed by the PDF generation process. Old September standalone core or AXI overlay results are not used as evidence for this complete October design.